SAUVI: AN IN-DEPTH VIETNAMESE SPOKEN LANGUAGE UNDERSTANDING AND REASONING BENCHMARK

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ABSTRACT

Recent advances in Large Audio Language Models (LALMs) have been driven in large part by increasingly sophisticated evaluation benchmarks. However, while dominant languages benefit from linguistically grounded evaluation suites, lower-resourced languages like Vietnamese remain severely underserved, often limited to basic perception tasks. To bridge this gap, we introduce SAUVI, a pioneering comprehensive benchmark designed for in-depth Vietnamese spoken language understanding and reasoning. SAUVI combines broad coverage of conventional capabilities and specialized tasks targeting Vietnamese linguistic phenomena, regional dialects, ethnic acoustics, and cultural auditory landscapes, resulting in 27 distinct tasks across both speech and music. Through a rigorous evaluation of state-of-the-art LALMs, we reveal key insights to guide future LALM development for under-represented languages.

***Index Terms*—** Large audio-language model, Multi-hop reasoning, Vietnamese, Benchmark

1. INTRODUCTION

The landscape of speech understanding has witnessed a remarkable paradigm shift in recent years, driven by advances in large language models, evolving from task-specific models into unified highly capable Large Audio Language Models (LALMs) [1, 2, 3].

This rapid acceleration in modeling capabilities is fundamentally driven by the continuous development of comprehensive evaluation benchmarks. Evaluation benchmarks have evolved significantly, progressing from foundational perception tasks [4, 5] toward complex reasoning and knowledge extraction [6], and increasingly targeting advanced cognitive skills [7, 8]. As the field pushes toward increasingly sophisticated reasoning capabilities, we argue that evaluation should also extend across linguistic boundaries to capture the unique characteristics of individual languages.

While dominant languages have established benchmarks grounded in language-specific phenomena, such as MMSU [7] for English linguistic phenomena or CMDAR [9] for dynamic Chinese audio reasoning, lower-resourced languages like Vietnamese still lack comprehensive evaluation suites. Although recent efforts like [10] introduce speech-understanding evaluations for Southeast Asian languages, including Vietnamese, they remain restricted to speech-based audio and are limited to a few general tasks (e.g., ASR and gender recognition). Given that Vietnamese is a richly characteristic language shaped by a profound history and rich cultural and linguistic diversity, a rigorous evaluation calls for a holistic approach that extends far beyond surface-level metrics.

To bridge this gap, we introduce **SAUVI** (In-depth Vietnamese Spoken Language Understanding and Reasoning Benchmark), a pioneering comprehensive evaluation framework tailored to Vietnamese. SAUVI is distinguished by two key features: (1) **Broad Conventional Coverage**: it provides broad evaluation of standard capabilities across both perception and reasoning; and (2) **Vietnamese-specific Expansion**: it goes beyond conventional benchmarks by introducing specialized tasks targeting distinctive linguistic phenomena, ethnic minority acoustics, and regional dialects. In total, SAUVI encompasses 27 tasks spanning both speech and music. Through thorough evaluations across different LALMs, we identify key challenges and provide insights to guide future development of LALMs for the Vietnamese language. Relevant resources can be found at `https://github.com/thuantn210823/SAUVI`.

2. THE SAUVI BENCHMARK

2.1. Benchmark Overview

Moving beyond conventional evaluations, SAUVI expands the assessment scope by explicitly targeting the intrinsic linguistic and cultural characteristics of the Vietnamese language. The benchmark features 16,894 high-quality, expert-annotated questions spanning two major audio domains, speech and music, which are distributed across 10 percep-

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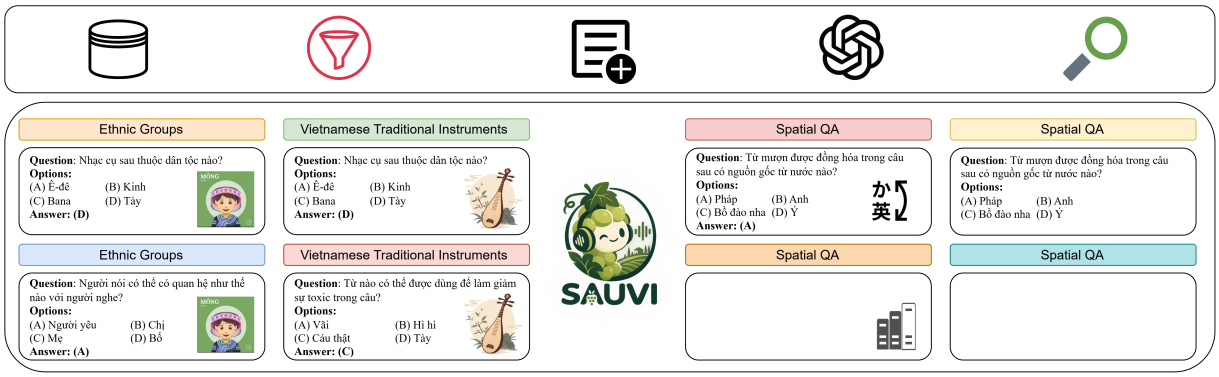


Fig. 1. Overview of the proposed FLOW-TS framework.

Table 1. Source datasets used in the benchmark. Task notions are shown in Fig. 2

Name	Tasks	License
VNTM [?]	P9, R16	None†
VietLyrics [?]	P7, R16	None†
ViMD [?]	P1–2, P8, R3–5, R7–12, R14, R18	CC BY-NC-ND 4.0
GigaSpeech2-VI [?]	P6, R3, R5–12, R14, R18	Apache-2.0‡
Sentiment-Reasoning [?]	P5, R14, R18	MIT
ViSEC [?]	P4, R4	CC BY 4.0
Speech-MASSIVE [?]	P2–3, R3–5, R7, R9–10, R12, R14, R18	CC BY-NC-SA 4.0
MultiMed-ST [?]	R1, R12, R18	MIT
VietMed-Sum [?]	R2, R14	None†
ViToSA [?]	R5, R7–10, R12–14, R18	CC BY-NC-ND 4.0
VietMDD [?]	R7, R10, R15	None†
VietMed [?]	R4–5, R7–10, R18	MIT
VLSP 2020 [?]	R3, R5, R7, R9–12, R18	VLSP DUA
VLSP 2021 Task 2 [?]	R5, R7, R9–11, R14, R18	VLSP DUA
VietMed-NER [?]	R3, R7	None†
ViMedCSS [?]	R3, R5, R7–12, R14, R18	CC BY 4.0
ASR-VLSP-2023 [?]	R3, R5, R7, R9–11, R13–14, R18	VLSP UA
VietNamNhac [?]	P9, R16–17	None†
VLSP 2021 Task 1 [?]	R3, R5, R7, R9–11, R18	VLSP DUA
VN-SLU [?]	R3, R5, R7–12	None†

† No explicit public license found. ‡ Non-commercial research/education only. DUA: Data User Agreement.

tion tasks and 20 reasoning tasks. To provide a nuanced evaluation, all questions are distinguished by difficulty, forming two distinct evaluation sets tailored for both standard and advanced capability assessments.

2.2. Data Construction

We construct SAUVI through a three-stage pipeline with rigorous quality control.

Tasks Design and Data Curation. To construct a comprehensive benchmark, we adopt a data-driven taxonomy design. We first aggregate existing speech and audio datasets, prioritizing human-annotated or well-validated test sets to ensure broad coverage of general audio-language capabilities. Vietnamese linguistic experts then inspect each dataset to audit annotation quality, assess task feasibility, and guarantee a well-rounded evaluation across diverse linguistic nuances. Based on this expert audit and dataset availability, the task taxonomy is iteratively refined, incorporating new tasks as distinct capabilities emerge and utilizing manual data curation where necessary.

Metadata and QA Construction. For each task, we construct the metadata required for scalable QA generation using existing annotations, automatic extraction, external knowledge sources, or LLM generation followed by verification. For knowledge-intensive tasks, related attributes are linked into lightweight knowledge graphs to support multi-hop reasoning. Expert-designed templates are then used to generate QA pairs whenever the reasoning structure can be explicitly defined; otherwise, LLMs generate questions following human-designed task and QA specifications. Throughout construction, we enforce audio dependency, following the principle *audio evidence* $\rightarrow$ *reasoning* $\rightarrow$ *answer*, and avoid questions solvable from textual shortcuts alone.

Quality Control. Generated QA pairs are first screened by LLMs for factual errors, ambiguity, multiple valid answers, weak distractors, and insufficient audio grounding. Flagged samples are manually verified and corrected or removed. Finally, experts review a subset of each task for correctness, naturalness, audio dependency, and distractor quality. Systematic issues are fed back to the metadata, templates, or generation instructions, and the corresponding data are regenerated until the task meets our quality criteria.

This process results in a quality-controlled benchmark covering both perception and reasoning tasks over speech and music.

2.3. Comparison and Task Coverage

Vietnamese is already considered in [10]. However, from a breadth perspective, it remains primarily speech-oriented, with limited coverage of perception and reasoning, whereas SAUVI expands the scope to 27 tasks spanning speech and music. From a depth perspective, [10] largely targets conventional capabilities, without systematically considering the

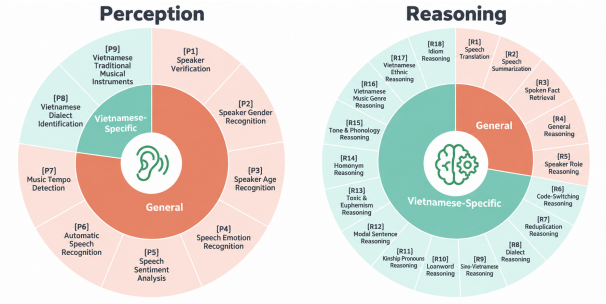


Fig. 2. Taxonomy of Perception and Reasoning tasks.

distinctive acoustic, linguistic, and cultural characteristics of Vietnamese. Figure 2 presents the SAUVI taxonomy, with perception and reasoning accounting for 57.09% and 42.91%, respectively. Among these, Vietnamese-specific tasks comprise 67% of the total, featuring reasoning tasks that may require multi-hop inference mirroring real-life speech comprehension.

For comprehensive evaluation, SAUVI includes broad conventional tasks [P1-P7][R1-R5], such as ASR and Speech Emotion Recognition for perception, coupled with cognitive reasoning skills like counting and temporal reasoning. However, to capture the true depth of Vietnamese, SAUVI introduces a specialized suite of localized tasks structured around three core dimensions:

- **Regional Diversity**: Vietnam's diverse geography gives rise to distinct regional varieties across its three regions and 63 provinces/cities (before 2026), such as pronunciation, phonetics, and vocabulary [P8], [R8].
- **Ethnic Diversity & Cultural Heritage**: Vietnam is home to 54 ethnic groups (e.g., Muong, Hmong) [R17], indigenous instrumentations (e.g., Dan bau) [P9], and culturally iconic genres (e.g., Ca tru singing) [R16].
- **Linguistic & Phonetic Complexity**: Vietnam's unique historical and cultural interactions have shaped distinctive linguistic and phonetic phenomena, including code-switching, reduplication, loanwords, and other language-specific patterns [R6–R18].

3. EXPERIMENTS

3.1. Evalution Setup

We comprehensively evaluate the proposed benchmark across a selection of state-of-the-art systems. Our evaluation suite includes Qwen-Audio-Chat [1], Qwen2-Audio-Instruct [2], and Kimi-Audio [3] as representatives of Large Audio Language Models (LALMs). The remaining baselines comprise Omni Large Language Models (OmniLLMs), including open-weight architectures such as Qwen2.5-Omni-7B [11], Gemma-4-E4B-it [12], Phi-4-Multimodal [13], and Qwen3-Omni-30B variants [11], as well as the commercial model Gemini-2.5-Flash[14].

Performance is measured using micro-averaged accuracy. Following the evaluation protocol proposed in [15], candidate responses generated by the LALMs are evaluated using an auxiliary LLM evaluator; specifically, we employ Gemini-2.5-Flash for this task. In instances where the LLM evaluator fails to produce a definitive judgment, the corresponding samples are manually reviewed and scored by human.

3.2. Results

3.2.1. Main results

Table 2 presents the evaluation results across all baselines. We summarize the key findings below:

Overall Performance and Model Architectures: Gemini-2.5-Flash and the open-weight Qwen3-Omni-30B-Thinking operate in a tier of their own, significantly outpacing the rest of the evaluation suite to establish the state-of-the-art on SAUVI. While the proprietary Gemini-2.5-Flash achieves the highest overall accuracy and excels in perception, Qwen3-Omni-30B-Thinking proves highly competitive by dominating all reasoning categories. Conversely, there is a severe capability cliff for smaller open-weight models. Systems such as Gemma-4-E4B-it, Qwen-Audio-Chat, and Phi-4-Multimodal completely fail to surpass the overall random chance baseline, demonstrating that the representational capacities of these smaller models remain heavily biased toward dominant languages and have not yet been adequately extended to handle Vietnamese.

General vs. Vietnamese-Specific Constraints: Generally, models exhibit strong proficiency on General tasks, with leading architectures approaching 70% accuracy. In stark contrast, performance deteriorates significantly on Vietnamese-specific tasks, where even the most capable models suffer accuracy drops exceeding 20%. This consistent disparity strongly validates the core premise of SAUVI: evaluating the true effectiveness of LALMs on a given language must extend far beyond conventional benchmarks to systematically probe localized linguistic, regional, and cultural nuances.

Perception vs. Reasoning Dynamics: An intriguing behavior emerges between cognitive levels: advanced models (e.g., Gemini-2.5-Flash, Qwen3-Omni-30B variants) paradoxically achieve higher scores in complex reasoning than in fundamental perception. This aligns with findings from English-centric benchmarks like MMSU [**?**] and MMAU [**?**], which note that current models suffer a fundamental deficiency in fine-grained acoustic perception despite strong semantic reasoning capabilities. We argue that if this perception bottleneck is already a critical gap for dominant languages, it demands even more rigorous attention for Vietnamese and

Table 2. Accuracies (%) and 95% confidence intervals of the baselines on SAUVI. The best and the second-best performances among open-source and proprietary LALMs are marked in bold and underlined.

Model	Size	Perception			Reasoning			Average
	(B)	General	Vietnamese	Average	General	Vietnamese	Average	All
Large Audio Language Models (LALMs)								
Qwen-Audio-Chat [1]	8.4B	27.71	21.07	23.92	18.03	19.36	19.04	21.82
Qwen2-Audio-Instruct [2]	8.4B	39.16	22.28	29.53	32.92	26.86	28.32	29.01
Kimi-Audio [3]	7B	48.71	26.95	36.29	55.50	39.37	43.27	39.29
Omni Large Language Models (OmniLLMs)								
Qwen2.5-Omni-7B [11]	7B	35.11	15.31	23.81	56.82	33.30	38.98	30.32
Gemma-4-E4B-it [12]	4B	25.11	13.31	21.81	36.82	23.30	28.98	20.32
Phi-4-Multimodal [13]	6B	28.70	19.74	23.59	27.27	20.71	22.29	23.03
Qwen3-Omni-30B-Instruct [11]	30B	61.51	31.49	44.38	67.20	49.73	53.95	48.48
Qwen3-Omni-30B-Thinking [11]	30B	**66.85**	35.25	48.81	**74.23**	**61.03**	**63.50**	55.11
Gemini-2.5-Flash [14]	-	64.46	**44.52**	**53.07**	70.18	55.46	58.21	**55.28**
Random Baseline								
Chance level	-	29.66	26.19	27.57	25.04	25.00	25.01	26.41

other lower-resourced languages, given their distinct acoustic and tonal complexities.

3.2.2. Task-specific analysis

4. CONCLUSION

In this paper, we presented SAUVI, the first comprehensive benchmark for evaluating Large Audio Language Models across 27 speech and music tasks tailored to the Vietnamese language and acoustic landscape. By combining broad conventional tasks with specialized evaluations of regional dialects, ethnic acoustics, and cultural phenomena, SAUVI exposes critical limitations of current LALMs and reveals a substantial performance gap between general capabilities and Vietnamese-specific nuances. Nevertheless, this gap is not uniform: frontier models perform remarkably well on complex reasoning tasks requiring deeper semantic understanding, particularly those involving Sino-Vietnamese words, loanwords, and modal sentences. This strong performance suggests an emerging ability to extend beyond dominant languages and engage with linguistic phenomena in less widely represented languages. However, this capability is less evident in smaller models, which struggle across nearly all task categories. SAUVI thus establishes a vital testbed for systematically measuring progress toward more capable and inclusive audio-language models, while encouraging the development of models that better capture the linguistic, acoustic, and cultural diversity of underrepresented languages.

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